

Welcome to Smart Agri Pack (SAP)

Innovating Agriculture with Smart Technology

At Smart Agri Pack (SAP), we specialize in delivering high-tech agricultural and commodity packaging machines designed for performance, precision, and productivity. With over 5 years of manufacturing expertise, SAP is committed to transforming traditional farming and packaging processes through advanced technology.

Whether you're packaging grains, nuts, spices, powders, oils, chips, or namkeen, our machines are built to handle it all — cleanly, efficiently, and reliably.

Global-Standard Technology

Our equipment is built with premium components imported from Germany, the UK, and China, ensuring international quality, durability, and cutting-edge automation.

Our Focus:

- Agri Commodity Packaging Machines
- Spices & Powder Filling
- Oil Pouch & Bottle Filling
- Snacks & Chips Packing Systems
- Custom Packaging Solutions for Food Industries

Why Choose SAP?

- 5+ Years of Industry Experience
- Export-Grade Machines with Global Reach
- Advanced Automation & PLC Control Systems
- Robust After-Sales Support

Join hundreds of satisfied clients across India and overseas who trust Smart Agri Pack for reliable, high-performance solutions in agriculture and food packaging.

Smart Technology. Smarter Farming.

Our fully-automated and semi-automated machines are engineered to meet the global standards of quality, efficiency, and reliability. All our advanced systems are powered by the latest technology **imported from Germany, UK, and China**, ensuring consistent performance and durability.

 Explore our range of smart packing solutions to boost productivity and reduce operational costs.

About Smart Agri Pack (SAP)

Innovating Agricultural Packaging for Over 5 Years

At Smart Agri Pack (SAP), we are dedicated to advancing the agricultural packaging industry through innovative technology and engineering excellence. Since our inception over 5 years

ago, we have been committed to providing high-quality, reliable, and efficient packing machines tailored for a wide range of agricultural commodities.

Our Mission

To empower farmers, food processors, and businesses with cutting-edge packaging solutions that enhance productivity, maintain product quality, and reduce waste.

What We Do

We design, manufacture, and export state-of-the-art packing machines specialized in handling:

- Grains & Pulses
- Dry Fruits & Nuts
- Spices & Spice Powders
- Edible Oils
- Chips & Namkeen
- Powders & Other Food Commodities

Our Technology

Our machines integrate advanced components and technology imported from Germany, the UK, and China, ensuring precision, durability, and compliance with international standards.

Why Choose SAP?

- Over 5 years of proven expertise in machine manufacturing
- Customized packaging solutions to meet diverse client needs
- Export-quality machinery trusted globally
- Strong after-sales support and service network

At Smart Agri Pack, we believe in combining innovation with reliability to help your business grow efficiently and sustainably.

Our Products

 **Advanced Farming Machines & Systems**

Smart Agri Pack (SAP) offers cutting-edge **advanced farming machines and systems** designed to optimize agricultural productivity and efficiency. Our range includes precision seeders, transplanters, soil preparation equipment, irrigation systems, harvesters, and smart farming technology — all engineered to meet the evolving needs of modern agriculture.

Fully Automated Packing Machines

We specialize in **fully automated packing machines** tailored for agricultural commodities and food products. Our machines deliver high-speed, accurate, and hygienic packaging solutions for spices, powders, grains, nuts, oils, chips, and snacks. Built with imported components from Germany, the UK, and China, our packing machines ensure minimal wastage, excellent sealing, and user-friendly operations with PLC and touch screen controls.

Services Page

Our Services

At Smart Tech Pack (SAP), we provide comprehensive packaging solutions for the agricultural and food processing sectors. Our services include:

Machine Manufacturing

We design and build high-performance packing machines for:

- Grains, Pulses & Cereals
- Nuts & Dry Fruits
- Spices (Whole & Ground)
- Powders (Atta, Maida, Turmeric, etc.)
- Edible Oil (Pouch & Bottle Filling)
- Chips & Namkeen Packaging

Machine Import & Integration

We source the latest machinery from **Germany, UK, and China** and adapt it to meet local compliance and customer requirements.

Maintenance & Support

Our experienced service engineers provide:

- Installation & Commissioning
- Annual Maintenance Contracts (AMC)
- Troubleshooting & Upgrades

Export Solutions

We export precision-engineered machines globally, built to match international standards and tailored to client needs.

Contact Us

Get in Touch with Smart Tech Pack (SAP)

Ready to upgrade your packing line? Let's talk!

 **Address:** [Your Company Address]

 **Phone:** +[Your Number]

 **Email:** info@smarttechpack.com

 **Website:** www.smartagripack.com

Working Hours:

Monday – Saturday: 9:00 AM – 6:00 PM

Sunday: Closed

Items to Pack with Grains, Pulses & Cereals Packing Machine

Grains:

- **Wheat**
- **Rice** (white, basmati, brown, parboiled)
- **Maize/Corn kernels**
- **Barley**
- **Millet** (Bajra, Ragi, etc.)
- **Sorghum** (Jowar)
- **Quinoa**

Pulses (Legumes):

- **Chickpeas** (Kabuli & Desi chana)
- **Lentils** (Red, Green, Yellow)
- **Green gram** (Moong dal)
- **Black gram** (Urad dal)
- **Pigeon peas** (Toor dal)
- **Kidney beans** (Rajma)
- **Black-eyed peas** (Lobia)
- **Split peas**

Cereals (Including breakfast cereals):

- **Oats** (rolled, instant)

- Cornflakes
- Wheat flakes
- Muesli
- Puffed rice (Murmura)
- Broken rice (Kanki)
- Semolina (Sooji/Rava)
- Cracked wheat (Dalia)

Types of Packaging Machines (Depending on Product Form & Weight)

- Automatic pouch packing machine
- Vertical Form Fill Seal (VFFS) machine
- Weigh filler or multi-head weigher
- Auger filler (for fine powder forms like flour, semolina)
- Cup filler (for uniform granules)
- Vacuum packing machine (for long shelf life)

Common Packaging Formats:

- Pouches (1kg, 500g, 100g)
- Plastic or laminated bags
- Stand-up pouches
- Zip-lock or resealable packs
- Paper or jute bags (for eco-friendly packaging)
- Bulk sacks (25kg, 50kg) for wholesale

Items to Pack with Nuts & Dry Fruits Packing Machine

Nuts:

- Almonds (whole, sliced, blanched)
- Cashew nuts (plain, roasted, salted)
- Peanuts (raw, roasted, salted, coated)
- Pistachios (shelled or in-shell)
- Walnuts (whole or kernels)
- Hazelnuts

- **Macadamia nuts**
- **Brazil nuts**
- **Pine nuts**

Dry Fruits:

- **Raisins (black, golden, green)**
- **Dried figs (Anjeer)**
- **Dates (Khajoor – dry & wet)**
- **Dried apricots**
- **Dried cranberries**
- **Dried blueberries**
- **Dried mango slices**
- **Dried banana chips**
- **Dried apple rings**

Mixed Products:

- **Trail mix**
- **Dry fruit mix**
- **Nut & seed blend**
- **Flavored dry fruits (masala, sweetened, etc.)**



Types of Packing Machines for Nuts & Dry Fruits

- **Automatic Weighing and Packing Machines**
- **Vertical Form Fill Seal (VFFS) Machines**
- **Multihead Weigher Machines** (high accuracy for uneven sizes like nuts)
- **Pneumatic Packing Machines** (for delicate handling)
- **Vacuum Packing Machines** (to extend shelf life)
- **Nitrogen Flushing Machines** (to preserve freshness)
- **Zipper Pouch & Stand-up Pouch Fillers**



Common Packaging Formats:

Packaging Type	Size Range	Features
Pillow Pouch	50g – 1kg	Cost-effective, compact
Stand-up Zipper Pouch	100g – 1kg	Resealable, premium look
Vacuum Packs	100g – 500g	Long shelf life
Jar or Container Fill	250g – 1kg	Rigid packaging
Bulk Bags	5kg – 25kg (wholesale)	For retail distribution

□ Notes:

- **Dry fruits are hygroscopic**, so moisture control is crucial—often packed with **desiccants** or **nitrogen gas flushing**.
- **Nuts are high-value** and need **accurate weight control** to reduce losses.
- Machines must comply with **food safety standards (like FSSAI, HACCP, ISO, etc.)**.

✓ Items to Pack with Spices Packing Machine

Whole Spices:

- Cumin seeds (Jeera)
- Black pepper
- Cloves
- Cardamom (green & black)
- Mustard seeds
- Coriander seeds
- Fennel seeds (Saunf)
- Fenugreek seeds (Methi)
- Bay leaves
- Cinnamon sticks
- Star anise
- Nutmeg
- Dry red chilies
- Asafoetida (Hing blocks)

Ground Spices (Powdered):

- Turmeric powder (Haldi)

- Chili powder (Lal Mirch)
 - Coriander powder
 - Cumin powder
 - Garam masala
 - Black pepper powder
 - Dry ginger powder (Sonth)
 - Clove powder
 - Cardamom powder
 - Fennel powder
 - Curry powder
 - Mixed masalas (Chaat masala, Sambar masala, etc.)
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 Types of Spices Packing Machines

For Whole Spices:

- **Weigh Filler Machines**
- **Cup Filler Machines**
- **VFFS Machines with Linear Weighers**
- **Semi-Automatic Granule Packing Machines**

For Ground Spices:

- **Auger Filler Machines** (screw-based, ideal for powders)
 - **Pneumatic Powder Packing Machines**
 - **VFFS Machines with Auger Filling**
 - **Vacuum or Nitrogen Flushing Machines** (to preserve freshness and aroma)
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 Common Packaging Formats:

Format	Suitable For	Features
Laminated pouches	Whole & ground	Airtight, customizable sizes
Zip-lock pouches	Ground spices	Reusable, retail-friendly
Sachets (small)	Sample packs (5–20g)	Economical for free trials

Format	Suitable For	Features
Spice jars (PET/HDPE)	Ground or blended	Premium, durable presentation
Stand-up pouches	Retail spice packs	Shelf display, better branding
Vacuum/foil pouches	High-aroma spices	Long shelf life, no moisture

Key Features to Look for in Spice Packing Machines:

- **Dust-proof design** (especially for powders)
- **Stainless steel contact parts (SS304 or SS316)**
- **Accurate dosing mechanism** (to avoid overfilling)
- **Heat sealing & date coding** options
- **Nitrogen flushing** for freshness and color retention

Items to Pack with Powder Packing Machines

Flours:

- **Wheat flour (Atta)**
- **Refined flour (Maida)**
- **Rice flour**
- **Gram flour (Besan)**
- **Ragi flour**
- **Corn flour**
- **Sooji (Semolina)**
- **Multigrain flour**

Spice Powders:

- **Turmeric powder (Haldi)**
- **Chili powder**
- **Coriander powder**
- **Cumin powder**
- **Black pepper powder**
- **Dry ginger powder**
- **Garam masala**

- **Chaat masala**
 - **Other Powders:**
 - **Salt**
 - **Milk powder**
 - **Protein powder**
 - **Coffee powder**
 - **Detergent powder** (non-food industrial use)
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 Types of Powder Packing Machines

 Auger Filler Packing Machine

- Most commonly used for all fine powders
- Uses a screw (auger) to fill precise quantities
- Can be **automatic**, **semi-automatic**, or **servo-based**

 VFFS (Vertical Form Fill Seal) with Auger Filler

- For making **pouches** from roll film
- Integrates **filling, sealing, and cutting** in one system

 Pneumatic Powder Packing Machine

- Higher speed and accuracy
- Great for commercial-grade turmeric, atta, or besan packing

 Manual or Semi-Automatic Packing Machines

- For small-scale businesses or low-volume packing
 - Cost-effective and easy to operate
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 Common Packaging Formats

Packaging Type	Weight Range	Best For
Laminated pouches	100g – 2kg	Retail packs (atta, spices)
Zip-lock pouches	250g – 1kg	Premium or export products
Paper/poly bags	5kg – 50kg	Wholesale atta, maida
Plastic jars/containers	100g – 500g	Spice powders or milk powder

Packaging Type	Weight Range	Best For
Sachets (mini)	10g – 50g	Samples, single-use spices

Key Features to Look For:

- **Dust control system** (essential for powder handling)
- **Stainless steel body (SS304/SS316)** for food safety
- **Nitrogen flushing** (for spice powders)
- **Automatic weight control** (load cell-based or servo)
- **Heat sealing** with batch/date printing
- **Multiple pouch sizes** with quick changeover

Types of Edible Oil Packing Machines

◆ 1. Pouch Filling Machines (Form-Fill-Seal - FFS):

- Fully automatic: forms pouch, fills oil, and seals it
- Capacity: 100 ml to 1 liter per pouch
- Suitable for **low-cost retail packs**
- Often includes **nitrogen flushing** and **date/batch coding**

◆ 2. Bottle Filling Machines:

- For PET, HDPE, glass bottles
- Can be **volumetric, servo, or gear pump-based**
- Capacities: 200 ml to 5 liters
- Often includes:
 - **Capping machines**
 - **Labeling units**
 - **Shrink sleeve applicators**

◆ 3. Jar & Tin Filling Machines:

- For 5L, 15L, 20L jars or tins
- Ideal for bulk or commercial packing
- May include **weigh-metric filling systems**

Packaging Formats

Packaging Type	Volume Range	Notes
Pouch (LDPE)	100ml – 1 liter	Economical, high-speed packing
PET/HDPE Bottles	200ml – 5 liters	Rigid, consumer-friendly, safe
Jerry Cans (plastic)	5L – 20L	Bulk or food service use
Metal tins	15L – 20L	Traditional for mustard/coconut oil

Key Machine Features to Look For

- **Drip-free nozzles** (no spillage or wastage)
- **SS316 contact parts** (food-grade)
- **Anti-leak heat sealing** for pouches
- **Adjustable filling volumes**
- **Automatic capping & labeling** (for bottles)
- **Nitrogen flushing system** (to preserve freshness)
- **Batch coding & QR/barcode printing** options

Ideal Machine Types by Business Size:

Business Type	Recommended Machine
Small/Startup	Semi-automatic bottle/pouch filler
Mid-scale (1–5 tons/day)	Automatic volumetric or gear pump filler
Large-scale plant	Servo-based multi-head linear fillers + pouch line

Chips & Namkeen Packaging Machines

Crisp Precision. Crunchy Results.

At **Smart Agri Pack**, we design advanced **Chips and Namkeen Packaging Machines** engineered for high-speed, hygienic, and accurate packing of all types of snacks.

Our machines are ideal for:

- Potato Chips
- Banana Chips

- Bhujia
- Sev
- Mixtures
- Fryums
- Puffed Snacks & More

✓ Key Features:

- **High-Speed Automatic Filling & Sealing**
- **Accurate Weighing with Multihead Weighers**
- **Nitrogen Flushing for Extended Shelf Life**
- **Leak-Proof Sealing with Zero Wastage**
- **Compatible with Pillow Pouch, Gusset, and Stand-Up Pouch Formats**
- **User-Friendly Touch Panel & PLC Control**

Built using **premium components sourced from Germany, UK & China**, our snack packing solutions ensure longer machine life, easy maintenance, and consistent output — making your brand stand out on shelves.

✓ White Rice Production Process (Step-by-Step)

1. Pre-cleaning

- Removes large impurities (stones, straw, dust)
- Machine: **Pre-cleaner / Vibrating sieve**

2. Dehusking (Husking)

- Removes the outer **husk** from the paddy
- Output: **Brown rice**
- Machine: **Paddy Husker / Rubber Roll Sheller**

3. Paddy Separator

- Separates **unhusked paddy** from **brown rice**
- Unhusked grains are returned to husker
- Machine: **Paddy Separator**

4. Polishing / Whitening

- Removes the **bran layer** to produce **white rice**
- Multiple polishing stages may be used for higher quality
- Machine: **Rice Whitener / Polisher**

5. Grading / Sorting

- Removes broken, short, or discolored grains
- Ensures uniform quality and size
- Machines:
 - **Rice Grader**
 - **Color Sorter** (optical sorting for export-grade rice)

6. Weighing & Packing

- Final clean white rice is packed for sale
- Machine: **Weigh Filler + Packing Machine**

Essential Rice Mill Machines List

Machine Name	Function
Pre-Cleaner	Removes impurities from paddy
Destoner	Removes small stones
Paddy Husker	Removes husk, produces brown rice
Paddy Separator	Separates brown rice from husk
Rice Whitener	Removes bran to make white rice
Rice Polisher	Gives glossy finish
Rice Grader	Separates broken/full rice
Color Sorter	Removes discolored grains
Packing Machine	Weights and packs final product

Rice Mill Plant Capacities

Plant Size	Capacity (per hour)	Ideal For
Mini Rice Mill	300 – 600 kg/hr	Small business/farmers
Small Scale	1 – 2 tons/hr	Local market supply
Medium Scale	3 – 4 tons/hr	Commercial processing units
Large Scale	5 – 10+ tons/hr	Export-level, high automation

Automation Options

- **Semi-Automatic:** Cheaper, manual loading/unloading
- **Fully Automatic:** Higher cost but labor-saving and faster
- **Turnkey Plants:** Complete setup including installation, training, and layout

Investment Estimate (Indicative Only)

Plant Type	Approx. Cost (INR)
Mini Mill (500kg/hr)	Rs 7,50000 – 12,50000
1 TPH Plant	\$22,50000 – \$37,50000
5 TPH Full Auto	\$85,00000 – \$150,000+

Advanced Farming Machine List

1. Precision Seeders & Planters

- GPS-guided seed drills
- Pneumatic planters
- No-till seeders
- Multi-crop planters

2. Transplanting Machines

- Vegetable transplanters
- Paddy transplanters (manual & automatic)
- Nursery seedling machines

3. Irrigation Equipment

- Drip irrigation systems
- Pivot irrigation machines
- Solar-powered irrigation pumps

- Automatic fertigation units

4. Harvesting Machines

- Combine harvesters
- Sugarcane harvesters
- Potato/Onion harvesters
- Forage harvesters

5. Crop Protection & Spraying Machines

- Drone sprayers (agri drones)
- Boom sprayers
- Mist blowers
- Tractor-mounted sprayers
- Battery-operated sprayers

6. Soil Preparation & Tillage Equipment

- Rotary tillers (rotavators)
- Power harrows
- Cultivators
- Subsoilers
- Laser land levelers

7. Post-Harvest Machines

- Grain dryers
- Cleaning & grading machines
- De-stoners
- Seed treaters
- Bagging & packing machines

8. Agri Commodity Packing Machines (by Smart Agri Pack)

- Spice & powder packaging machines
- Chips & snack packing machines
- Pulses & grain filling systems
- Oil pouch/bottle filling machines
- Vacuum & nitrogen flush packaging

9. Smart Farming & IoT Devices

- Soil moisture & pH sensors
- Weather monitoring stations
- Smart irrigation controllers
- GPS tractor auto-steering systems
- Crop health monitoring drones

10. Livestock & Dairy Automation

- Automatic milking machines
- Cattle feed mixers
- Cow dung separators
- Poultry battery cages & feeders